V93000 Specifications

V93000 Wave Scale RF RFIM-18

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Wave Scale RF with RFIM-18 specifications

In this section, Wave Scale RF is either WSRF-6 or WSRF-8 card.

Note The specifications provided here are preliminary and are subject to change without notice.

Note The temperature of the water circulating in the secondary cooling loop must be set to +30 # to guarantee the specified performance. For more details, see *Water specification secondary cooling loop of STH*.

Terms and conditions

spec	Specifications describe warranted product performance.
typ	Characteristics are included as typical values describe a performance that is not covered by the product warranty. They describe a performance that is met by 80 % of the units with a 95 % confidence level.
meas	Measured values characterize expected product performance by means of measurements provided on individual samples.

Wave Scale RF with RFIM-18 stimulus

RFIM-18 stimulus maximum output power

Shows absolute maximum output power, measured as saturated power.

Conditions: LO source WSRF. Measured in loopback. Receiver and mix receiver path direct. Programmed output power 25 dBm.

Carrier frequency (GHz)	Maximum output power (dBm) Spec
5.85 <= f < 13.75	10
13.75 <= f < 15.75	6
15.75 <= f < 17.75	10
17.75 <= f <= 18	9

RFIM-18 stimulus power accuracy

Conditions:

- Output power: -10 dBm
- Optimization: noisefloor
- Valid only when driving the stimulus into a 50 Ohm matched power sensor directly attached to the pogo. Any further cabling and/or mismatch will induce additional error.
- Option: powerAccuracyOptimization

Characteristic stimulus power accuracy (dB)

Power range	Frequency range (GHz)												
	5.85 <= f < 6	6 <= f < 7	7 <= f < 8	8 <= f < 9	9 <= f < 10	10 <= f < 11	11 <= f < 12	12 <= f < 13	13 <= f < 14	14 <= f < 15	15 <= f < 16	16 <= f < 17	17 <= f < 18
-80 <= Lv < -70	±1.2	±1.3	±1.4	±1.6	±1.6	±1.3	±1.0	±1.2	±1.3	±1.4	±1.4	±1.5	±1.4
-70 <= Lv < -60	±0.7	±1.0	±1.1	±1.3	±1.3	±1.1	±1.0	±1.0	±1.1	±1.1	±1.1	±1.4	±1.2
-60 <= Lv < -50	±0.7	±0.8	±0.9	±1.1	±1.1	±1.0	±0.9	±1.0	±1.0	±1.1	±1.1	±1.3	±1.2
-50 <= Lv < -40	±0.6	±0.8	±0.9	±1.1	±1.0	±1.0	±0.9	±0.9	±1.0	±1.0	±1.1	±1.2	±1.1
-40 <= Lv < -30	±0.6	±0.8	±0.9	±1.0	±1.0	±1.0	±0.9	±0.9	±1.0	±1.0	±1.1	±1.2	±1.1

Power range	Frequency range (GHz)												
	5.85 <= f < 6	**6 <= f < 7**	**7 <= f < 8**	**8 <= f < 9**	**9 <= f < 10**	**10 <= f < 11**	**11 <= f < 12**	**12 <= f < 13**	**13 <= f < 14**	**14 <= f < 15**	**15 <= f < 16**	**16 <= f < 17**	**17 <= f < 18**
-30 <= Lv < -20	±0.6	±0.8	±0.9	±0.9	±0.9	±0.9	±0.8	±0.9	±1.0	±1.0	±1.0	±1.1	±1.1
-20 <= Lv < -10	±0.6	±0.7	±0.8	±0.8	±0.7	±0.9	±0.7	±0.7	±0.8	±0.9	±0.9	±0.9	±0.9
-10 <= Lv < -8	±0.7	±0.7	±0.8	±0.8	±0.7	±0.7	±0.7	±0.6	±0.7	±0.7	±0.7	±0.9	±0.9
-8 <= Lv < -6	±0.7	±0.7	±0.8	±0.8	±0.8	±0.7	±0.7	±0.6	±0.7	±0.7	±0.7	±0.9	±0.9
-6 <= Lv < -4	±0.7	±0.7	±0.8	±0.8	±0.8	±0.7	±0.7	±0.6	±0.7	±0.8	±0.7	±0.9	±0.9
-4 <= Lv < -2	±0.7	±0.8	±0.8	±0.8	±0.8	±0.7	±0.7	±0.6	±0.8	±0.9	±0.8	±0.9	±0.9
-2 <= Lv < 0	±0.7	±0.8	±0.9	±0.8	±0.8	±0.7	±0.8	±0.7	±0.9	±1.0	±1.0	±0.9	±1.0
0 <= Lv < 2	±0.7	±0.8	±0.9	±0.9	±0.8	±0.7	±0.9	±0.7	±1.0	±1.1	±1.0	±1.0	±1.1
2 <= Lv < 4	±0.7	±0.8	±0.9	±0.9	±0.9	±0.8	±1.0	±0.8	±1.1	±1.4	±1.3	±1.1	±1.2
4 <= Lv < 6	±0.7	±0.8	±0.9	±0.9	±0.9	±0.9	±1.3	±1.0	±1.3				
6 <= Lv < 8	±0.7	±0.8	±0.9	±0.9	±1.0	±1.1	±1.3	±1.2					
8 <= Lv < 10	±0.8	±0.9	±1.0	±1.0	±1.1	±1.1							

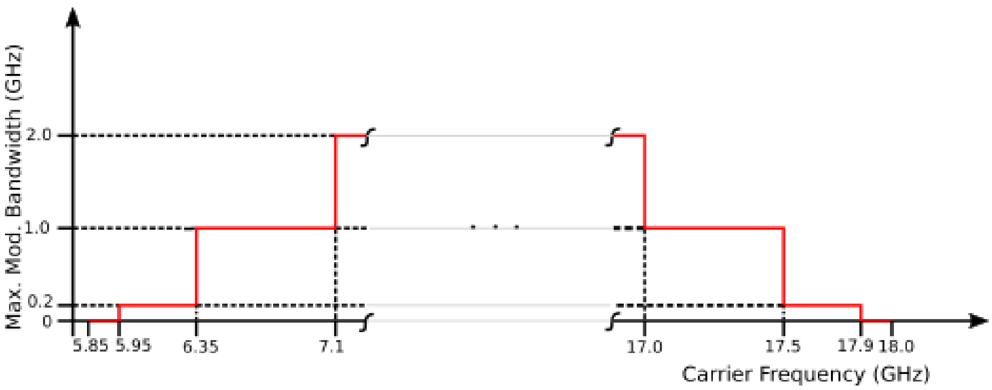
Typical stimulus power accuracy (dB)

Power range	Frequency range (GHz)												
	5.85 <= f < 6	**6 <= f < 7**	**7 <= f < 8**	**8 <= f < 9**	**9 <= f < 10**	**10 <= f < 11**	**11 <= f < 12**	**12 <= f < 13**	**13 <= f < 14**	**14 <= f < 15**	**15 <= f < 16**	**16 <= f < 17**	**17 <= f < 18**
-80 <= Lv < -70	±0.8	±0.8	±0.9	±1.1	±1.1	±0.9	±0.9	±0.8	±0.9	±0.9	±0.9	±1.0	±0.9
-70 <= Lv < -60	±0.6	±0.7	±0.8	±1.0	±1.0	±0.8	±0.7	±0.8	±0.8	±0.8	±0.8	±1.0	±0.8
-60 <= Lv < -50	±0.5	±0.6	±0.7	±0.8	±0.8	±0.7	±0.7	±0.7	±0.8	±0.8	±0.8	±0.9	±0.8
-50 <= Lv < -40	±0.5	±0.6	±0.7	±0.8	±0.8	±0.7	±0.7	±0.7	±0.8	±0.7	±0.8	±0.9	±0.8
-40 <= Lv < -30	±0.5	±0.6	±0.7	±0.8	±0.7	±0.7	±0.7	±0.7	±0.7	±0.7	±0.8	±0.9	±0.8
-30 <= Lv < -20	±0.5	±0.6	±0.7	±0.7	±0.6	±0.7	±0.7	±0.6	±0.7	±0.7	±0.7	±0.8	±0.7
-20 <= Lv < -10	±0.5	±0.5	±0.6	±0.6	±0.6	±0.6	±0.6	±0.5	±0.6	±0.7	±0.6	±0.6	±0.6
-10 <= Lv < -8	±0.5	±0.5	±0.6	±0.6	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.6	±0.6
-8 <= Lv < -6	±0.5	±0.5	±0.6	±0.6	±0.6	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.6	±0.6
-6 <= Lv < -4	±0.5	±0.5	±0.6	±0.7	±0.6	±0.5	±0.5	±0.5	±0.6	±0.6	±0.5	±0.6	±0.6
-4 <= Lv < -2	±0.5	±0.6	±0.6	±0.7	±0.6	±0.5	±0.5	±0.5	±0.6	±0.7	±0.6	±0.7	±0.7
-2 <= Lv < 0	±0.5	±0.6	±0.7	±0.7	±0.6	±0.5	±0.5	±0.5	±0.7	±0.8	±0.7	±0.7	±0.7

Power range	Frequency range (GHz)												
	5.85 <= f < 6	**6 <= f < 7**	**7 <= f < 8**	**8 <= f < 9**	**9 <= f < 10**	**10 <= f < 11**	**11 <= f < 12**	**12 <= f < 13**	**13 <= f < 14**	**14 <= f < 15**	**15 <= f < 16**	**16 <= f < 17**	**17 <= f < 18**
0 <= Lv < 2	±0.5	±0.6	±0.7	±0.7	±0.6	±0.5	±0.6	±0.6	±0.8	±0.8	±0.8	±0.8	±0.8
2 <= Lv < 4	±0.5	±0.6	±0.7	±0.7	±0.6	±0.6	±0.7	±0.7	±0.9	±1.1	±1.1	±0.8	±0.9
4 <= Lv < 6	±0.5	±0.6	±0.7	±0.7	±0.7	±0.7	±0.8	±0.8	±1.0				
6 <= Lv < 8	±0.5	±0.6	±0.7	±0.7	±0.8	±0.8	±1.0	±0.9					
8 <= Lv < 10	±0.6	±0.7	±0.7	±0.7	±0.9	±0.9							

RFIM-18 stimulus modulation bandwidth

To meet the RFIM-18 specifications, the restrictions on a modulated stimulus are: (Carrier Frequency - Mod. Bandwidth/2) >= 5.85 GHz. However, SmarTest only checks carrier frequency against 5.2 GHz. If you use a modulated signal at a carrier frequency of 5.2 GHz, the result will be attenuated by a filter.



Stim Mod. Bandwidth vs. Carrier Frequency

RFIM-18 stimulus phase noise

CW stimulus phase noise (dBc/Hz), measured at 0 dBm stimulus power. Optimization is "noiseFloor".

LO source WSRF stimulus phase noise

Frequency (GHz)	Characteristic stimulus phase noise (dBc/Hz) Offset frequency				
	1 kHz	**10 kHz**	**100 kHz**	**1 MHz**	**10 MHz**
6	-80.8	-94.9	-104.5	-104.5	-129.3
18	-75.6	-89.9	-98.0	-100.8	-129.3

LO source WSPS stimulus phase noise

Frequency (GHz)	Characteristic stimulus phase noise (dBc/Hz) Offset frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
6	-83.6	-101.1	-109.7	-112.1	-129.4
18	-85.8	-104.8	-108.8	-115.5	-130.0

RFIM-18 stimulus TOI (third-order intercept)

Conditions: Tone power is 0 dBm, Optimization is "imd", SmT source is "modulated".

WSRF base band 100 kHz spacing

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
5.2	22	26
6	22	25
7.2	20	23
8	21	24
9.2	21	25
10	17	21
11.2	16	19
12	14	18
13.2	14	18
14	13	16
15.2	9	12
16	14	19
17.2	15	20
18	11	16

WSRF base band 40 MHz spacing

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
5.2	21	25
6	22	25
7.2	22	25
8	23	26
9.2	20	25
10	17	21
11.2	15	18
12	14	18
13.2	13	16

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
14	12	15
15.2	8	12
16	14	18
17.2	16	20
17.96	11	16

AUX base band 100 kHz spacing

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
5.2	22	26
6	23	26
7.2	20	25
8	21	25
9.2	18	25
10	17	22
11.2	18	22
12	17	22
13.2	16	19
14	15	18
15.2	10	13
16	15	20
17.2	16	21
18	12	17

AUX base band 40 MHz spacing

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
5.2	20	24
6	21	24
7.2	23	26
8	23	26
9.2	21	26
10	16	21
11.2	16	20
12	15	19
13.2	15	18
14	13	16
15.2	9	12

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
16	15	19
17.2	17	21
17.96	13	18

RFIM-18 stimulus spurs

Second harmonic spurs

Conditions: LO source WSRF or WSPS. Programmed output power 0 dBm.

WSRF path

Carrier frequency (GHz)	Max spur (dBc) Characteristic	Max spur (dBc) Typ
5.2	-34.7	-38.0
5.85	-34.9	-36.8
9		
12		
15		
18		

AUX path

Carrier frequency (GHz)	Max spur (dBc) Characteristic	Max spur (dBc) Typ
5.2	-26.8	-36.2
5.85	-33.1	-36.4
9		
12		
15		
18		

Stimulus LO feedthrough spurs

Conditions: LO source WSRF. Programmed output power 0 dBm.

WSRF path

Frequency (GHz)	Max spurs (dBc) Characteristic	Max spurs (dBc) Typ
5.2	-2.4	-5.1
5.85	-37.3	-47.9
9		

Frequency (GHz)	Max spurs (dBc) Characteristic	Max spurs (dBc) Typ
12		
15		
18		

AUX path

Frequency (GHz)	Max spurs (dBc) Characteristic	Max spurs (dBc) Typ
5.2	-7.3	-17.8
5.85	-37.9	-48.3
9		
12		
15		
18		

Unknown spurs

Conditions: LO source WSRF. Programmed output power 0 dBm.

From the result of the unknown spurs, the known ones are removed. Known spurs are considered the following:

- (RF x m) + (BB x n)
- abs((RF x m) - (BB x n))
- (BB x n) + (LO x p)
- abs((BB x n) - (LO x p))
- (LO x p) + (RF x m) and abs((LO x p) - (RF x m))

using the following variables:

- m: 0, 1, 2, 3, 4, 5
- n: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- p: 0, 1/4, 1/2, 3/4, 1, 2, 3

WSRF path

Frequency (GHz)	Max spurs (dBc) Characteristic	Max spurs (dBc) Typ
5.2	-47.1	-50.9
5.85	-43.5	-53.2
9		
12		
15		
18		

AUX path

Frequency (GHz)	Max spurs (dBc) Characteristic	Max spurs (dBc) Typ
5.2	-44.6	-51.5
5.85	-44.6	-54.0
9		
12		
15		
18		

RFIM-18 stimulus ACLR (adjacent channel leakage ratio)

5GNR 1Carrier ACLR, stim power: -15 dBm to 0 dBm

WSRF path

Frequency (GHz)	ACLR1 (dBc) Characteristic	ACLR1 (dBc) Typ	ACLR2 (dBc) Characteristic	ACLR2 (dBc) Typ
10	-33	-37	-45	-49
14	-34	-37	-46	-49
17.5	-34	-37	-43	-48

Optimizer: balanced, NoiseBandwidth = 100 MHz

AUX path

Frequency (GHz)	ACLR1 (dBc) Characteristic	ACLR1 (dBc) Typ	ACLR2 (dBc) Characteristic	ACLR2 (dBc) Typ
10	-34	-37	-48	-52
14	-33	-37	-46	-52
17.5	-34	-38	-45	-50

Optimizer: balanced, NoiseBandwidth = 100 MHz

RFIM-18 stimulus EVM (error vector magnitude)

Conditions:

- Signal format: 5GNR
- Signal bandwidth: 100 MHz

LO source WSRF, Stim path WSRF

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-5	1.94	1.59
14	-5	2.17	1.75
17.5	-5	2.45	2.02

Optimizer: balanced, NoiseBandwidth= 5 MHz

LO Source WSPS, Stim path WSRF

Frequency (GHz)	**Signal Power (dBm)**	**EVM (%) Characteristic**	**EVM (%) Typ**
10	-5	1.62	1.14
14	-5	1.61	1.32
17.5	-5	1.71	1.45

Optimizer: balanced, NoiseBandwidth= 5 MHz

LO source WSRF, Stim path AUX

Frequency (GHz)	**Signal Power (dBm)**	**EVM (%) Characteristic**	**EVM (%) Typ**
10	-5	1.72	1.48
14	-5	1.98	1.73
17.5	-5	2.12	1.84

Optimizer: balanced, NoiseBandwidth= 5 MHz

LO source WSPS, Stim path AUX

Frequency (GHz)	**Signal Power (dBm)**	**EVM (%) Characteristic**	**EVM (%) Typ**
10	-5	1.12	0.97
14	-5	1.53	1.3
17.5	-5	1.37	1.22

Optimizer: balanced, NoiseBandwidth= 5 MHz

Wave Scale RF with RFIM-18 measure

RFIM-18 measure power

Conditions:

- WSRF receiver path: direct
- Actual input power setting (dBm): +25 (direct path), +15 (preamp path)
- SmarTest expected input power setting (dBm): +20 (direct path), +20 (preamp path)

Maximum RF input power to all ports

Mix receiver path	Receiver frequency range CW (GHz)	Maximum measurable power with 0 dB att. (dBm) Characteristic
direct	5.85 < = f < 12	17.5
	12 <= f < 13	17.0
	13 <= f < 17	16.5
	17 <= f <= 18	15.5
preamp	5.85 <= f <= 18	0.5

Absolute maximum power damage level (dBm)

- For direct path: 24
- For preamp path: 22

RFIM-18 measure power accuracy

Characteristic measure power accuracy - direct path

Conditions: Bandwidth of interest is set to 1 MHz. To achieve the specified level accuracy for power levels < -30dBm appropriate bandwidth of interest and result averaging must be applied.

Characteristic power accuracy (dB) direct path

Power range (dBm)	Frequency range		
	6 GHz <= 12 GHz	12 GHz <= 16 GHz	16 GHz <= 18 GHz
< -30	±0.40	±0.46	±0.49
-30 <= 0	±0.40	±0.46	±0.49
0 <= 10	±0.38	±0.48	±0.55
10 <= 14	±0.65	±0.72	±0.89 [1]

Characteristic measure power accuracy - preamp path

Conditions: Bandwidth of interest is set to 1 MHz. To achieve the specified level accuracy for power levels < -50dBm appropriate bandwidth of interest and result averaging must be applied.

[1] *17 GHz <= 18 GHz; 10 dBm <= 12 dBm*

Characteristic power accuracy (dB) preamp path

Power range (dBm)	Frequency range		
	6 GHz <= 12 GHz	12 GHz <= 16 GHz	16 GHz <= 18 GHz
< -50	±0.45	±0.53	±0.54
-50 <= -10	±0.45	±0.53	±0.54
-10 <= -6	±0.42	±0.56	±0.68
-6 <= -3	±0.57	±0.78	±0.81 [2]

Typical measure power accuracy - direct path

Conditions: Bandwidth of interest is set to 1 MHz. To achieve the specified level accuracy for power levels < -30dBm appropriate bandwidth of interest and result averaging must be applied.

Typical power accuracy (dB) direct path

Power range (dBm)	Frequency range		
	6 GHz <= 12 GHz	12 GHz <= 16 GHz	16 GHz <= 18 GHz
< -30	±0.29	±0.33	±0.40
-30 <= 0	±0.29	±0.33	±0.40
0 <= 10	±0.28	±0.36	±0.45
10 <= 14	±0.60	±0.64	±0.81[1]

Typical measure power accuracy - preamp path

Conditions: Bandwidth of interest is set to 1 MHz. To achieve the specified level accuracy for power levels < -50dBm appropriate bandwidth of interest and result averaging must be applied.

Typical power accuracy (dB) preamp path

Power range (dBm)	Frequency range		
	6 GHz <= 12 GHz	12 GHz <= 16 GHz	16 GHz <= 18 GHz
< -50	±0.34	±0.44	±0.44
-50 <= -10	±0.34	±0.44	±0.44
-10 <= -6	±0.33	±0.47	±0.57
-6 <= -3	±0.50	±0.70	±0.70[2]

RFIM-18 measure modulation flatness

The typical and characteristic performance for RFIM-18 measure modulation flatness is shown in the table below.

Conditions: RF power is 0 dBm, expected input power is +5 dBm, mix receiver path is direct, Bandwidth of interest (For WSRF numbers) is 210 MHz.

The flatness is measured using a fixed LO and a swept BB. The BB center frequency is set to default.

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[2] 16 GHz <= 17 GHz; -6 dBm <= -4 dBm   17 GHz <= 18 GHz; -6 dBm <= -3 dBm

Measure path	RF center frequency (GHz)	Maximum modulation bandwidth (GHz)	Modulation flatness (dB) Typ	Modulation flatness (dB) Characteristic
WSRF	6.5	0.2	1.7	3.7
	17.8	0.2		
AUX	6.5	1	4.9	9.1
	17.3	1		
	7.4	2	4.6	8.3
	8	2		
	8.9	2		
	10.5	2		
	12	2		
	14	2		
	16	2		
	16.8	2		

RFIM-18 measure noise floor

Conditions: Measured at 0 dB receiver attenuation using the internal termination. The Stim path is set to 18 GHz and -90 dBm.

Mix receiver path	Receiver frequency range CW (GHz)	Avg. noise floor (dBm/Hz) Spec
direct	5.2 <= f < 8.2	-144
	8.2 <= f < 11.2	-143
	11.2 <= f < 13.2	-142
	13.2 <= f < 15.6	-141
	15.6 <= f <= 18.0	-138
preamp	5.2 <= f < 8.2	-157
	8.2 <= f < 10.2	-156
	10.2 <= f < 13.0	-155
	13.0 <= f < 15.6	-154
	15.6 <= f <= 18.0	-153

RFIM-18 measure phase noise

CW measure phase noise (dBc/Hz), measured at 0 dBm RF power.

LO source WSRF measure phase noise (dBc/Hz)

Input carrier frequency (GHz)	Characteristic measure phase noise (dBc/Hz) Offset frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
6	-78.7	-95.5	-103.8	-105.6	-127.6

Input carrier frequency (GHz)	Characteristic measure phase noise (dBc/Hz) Offset frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
18	-72.6	-89.3	-96.1	-101.3	-127.1

LO source WSPS measure phase noise (dBc/Hz)

Input carrier frequency (GHz)	Characteristic measure phase noise (dBc/Hz) Offset frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
6	-80.4	-101.2	-109.4	-112.9	-128.0
18	-85.3	-103.0	-106.9	-113.2	-128.9

RFIM-18 measure two-tone intermodulation (IP3)

Conditions: Measured at -5 dBm for direct path and -20 dBm for preamp path. WSRF receiver path is preamp1. Frequency range 5.85 <= f <= 18 GHz.

WSRF path

Mix receiver path	100 kHz tone spacing TOI (dBm) Characteristic	40 MHz tone spacing TOI (dBm) Characteristic
direct	19.5	20.3
preamp	3.6	4.4

AUX path

Mix receiver path	100 kHz tone spacing TOI (dBm) Characteristic	40 MHz tone spacing TOI (dBm) Characteristic
direct	23.2	23.6
preamp	6.4	6.3

RFIM-18 measure spurs

WSRF baseband: valid for spurs frequency offset +/- 100 MHz

Frequency (GHz)	Max spur (dBc) Characteristic	Max spur (dBc) Typ
5.2	-23	-27
5.85	-49	-54
9	-46	-50
12	-54	-57
15	-42	-46
18	-42	-45

AUX baseband: valid for spurs frequency offset +/- 100 MHz

Frequency (GHz)	Max spur (dBc) Characteristic	Max spur (dBc) Typ
5.2	-41	-51
5.85	-44	-52
9	-41	-48
12	-44	-54
15	-34	-44
18	-36	-43

RFIM-18 measure ACLR (adjacent channel leakage ratio)

Conditions:

- LO source WSRF or WSPS
- Signal format: 5GNR
- SmarTest expected input power: Signal power + PeakToAvgPowerRatio
- SmarTest PeakToAvgPowerRatio(5GNR): 12.0 dB
- SmarTest resultAveraging: 8

WSRF path

Frequency (GHz)	Signal Power (dBm)	ACLR1 (dBc) Characteristic	ACLR1 (dBc) Typ
10	-28 <= P <= 0	-44.1	-45.8
14	-28 <= P <= 0	-44.0	-45.4
17.5	-28 <= P < -4	-40.4	-43.7
	-4 <= P <= 0	-38.3	-42.2

AUX path

Frequency (GHz)	Signal Power (dBm)	ACLR1 (dBc) Characteristic	ACLR1 (dBc) Typ
10	-28 <= P <= 0	-45.7	-48.1
14	-28 <= P <= 0	-42.6	-45.6
17.5	-28 <= P < -4	-44.6	-48.2
	-4 <= P <= 0	-41.9	-44.1

RFIM-18 measure EVM (error vector magnitude)

Conditions:

- Power range: -4 to -16 dBm for direct path and -16 to -28 dBm for preamp path.
- Signal format: 5GNR
- SmarTest Bandwidth of Interest: 100 MHz
- SmarTest expected input power: Signal power + PeakToAvgPowerRatio

- SmarTest PeakToAvgPowerRatio (5GNR): 12.0 dB

LO source WSRF, meas path WSRF

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-28 <= P <= -4	1.80	1.40
14	-28 <= P <= -4	2.00	1.55
17.5	-28 <= P <= -4	2.45	1.85

LO source WSPS, meas path WSRF

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-28 <= P <= -4	1.00	0.80
14	-28 <= P <= -4	1.00	0.80
17.5	-28 <= P <= -4	1.05	0.90

LO source WSRF, meas path AUX

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-28 <= P <= -4	1.80	1.50
14	-28 <= P <= -4	2.20	1.60
17.5	-28 <= P <= -4	2.40	1.85

LO source WSPS, meas path AUX

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-28 <= P <= -4	1.15	0.95
14	-28 <= P <= -4	1.20	1.00
17.5	-28 <= P <= -4	1.35	1.00

Wave Scale RF with RFIM-18 loopback

RFIM-18 loopback ACLR (adjacent channel leakage ratio)

5GNR, Uplink, 100 MHz bandwidth, HS ADC

Mix receiver path	Frequency (GHz)	Power range (dBm)	ACLR1 upper/ lower (dBc) Spec	ACLR2 upper/ lower (dBc) Spec
direct	10.0	-12 to -4	-39	-40
	14.0		-39	-40
	17.5		-38	-38
preamp	10.0	-20 to -12	-38	-40
	14.0		-39	-41
	17.5		-38	-39

Conditions: Optimizer: balanced, NoiseBandwidth = 75 MHz, baseband attenuation = 6dB

RFIM-18 loopback EVM (error vector magnitude)

LO source WSRF

5GNR, Uplink, 100 MHz bandwidth, HS ADC

Frequency (GHz)	Power (dBm)	EVM (%) Spec
10.0	-10	2.2
14.0		2.6
17.5		2.7

Optimizer: balanced, NoiseBandwidth= 5 MHz, baseband attenuation 0dB

LO source WSPS

5GNR, Uplink, 100 MHz bandwidth, HS ADC

Frequency (GHz)	Power (dBm)	EVM (%) Spec
10.0	-10	1.5
14.0		1.4
17.5		1.5

Optimizer: balanced, NoiseBandwidth= 5 MHz, baseband attenuation 0dB